

Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$\mathcal{K}_{\alpha\beta\chi}$	$\mathcal{K}_{\alpha\chi\beta} == -\mathcal{K}_{\alpha\beta\chi}$	$\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi} - \frac{1}{2} \eta_{\alpha\chi} \mathcal{K}_{1^-}^{\#1}{}_{\beta} + \frac{1}{2} \eta_{\alpha\beta} \mathcal{K}_{1^-}^{\#1}{}_{\chi} + \frac{4}{3} \mathcal{K}_{2^-}^{\#1}{}_{\beta\chi\alpha} + \mathcal{K}_{1^+}^{\#2}{}_{\beta\chi} n_{\alpha} +$ $\frac{1}{3} \eta_{\alpha\chi} \mathcal{K}_{0^+}^{\#1}{}_{n_{\beta}} - \mathcal{K}_{1^+}^{\#1}{}_{\alpha\chi} n_{\beta} + \mathcal{K}_{2^+}^{\#1}{}_{\alpha\chi} n_{\beta} - \frac{1}{2} \mathcal{K}_{1^-}^{\#1}{}_{\chi} n_{\alpha} n_{\beta} - \mathcal{K}_{1^-}^{\#2}{}_{\chi} n_{\alpha} n_{\beta} -$ $\frac{1}{3} \eta_{\alpha\beta} \mathcal{K}_{0^+}^{\#1}{}_{n_{\chi}} + \mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} n_{\chi} - \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} n_{\chi} + \frac{1}{2} \mathcal{K}_{1^-}^{\#1}{}_{\beta} n_{\alpha} n_{\chi} + \mathcal{K}_{1^-}^{\#2}{}_{\beta} n_{\alpha} n_{\chi}$	$\mathcal{I}_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$\mathcal{K}_{0^+}^{\#1}$	(None)	$\mathcal{K}_{\alpha\beta}^{\beta} n^{\alpha}$	$\mathcal{I}_{0^+}^{\#1}$
$\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$	$\mathcal{K}_{0^+}^{\#1}{}_{\alpha\chi\beta} == -\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ & $\mathcal{K}_{0^+}^{\#1}{}_{\beta\alpha\chi} == -\mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi}$	$\frac{1}{3} \mathcal{K}_{\alpha\beta\chi} - \frac{1}{3} \mathcal{K}_{\beta\alpha\chi} + \frac{1}{3} \mathcal{K}_{\chi\alpha\beta} - \frac{1}{3} \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\delta} +$ $\frac{1}{3} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\delta} - \frac{1}{3} \mathcal{K}_{\delta\beta\chi} n_{\alpha} n^{\delta} + \frac{1}{3} \mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\delta} +$ $\frac{1}{3} \mathcal{K}_{\delta\alpha\chi} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{K}_{\alpha\beta\delta} n_{\chi} n^{\delta} + \frac{1}{3} \mathcal{K}_{\beta\alpha\delta} n_{\chi} n^{\delta} - \frac{1}{3} \mathcal{K}_{\delta\alpha\beta} n_{\chi} n^{\delta}$	$\mathcal{I}_{0^+}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1}{}_{\beta\alpha} == -\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\frac{1}{2} \mathcal{K}_{\alpha\beta\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\beta\alpha\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}_{1^+}^{\#1}{}_{\alpha\beta}$
$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\beta\alpha} == -\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{\chi\alpha\beta} n^{\chi} + \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} - \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}_{1^+}^{\#2}{}_{\alpha\beta}$
$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	(None)	$-\mathcal{K}_{\alpha\beta}^{\beta} + \mathcal{K}_{\beta\chi}^{\chi} n_{\alpha} n^{\beta} + \mathcal{K}_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$\mathcal{I}_{1^+}^{\#1}{}_{\alpha}$
$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$	(None)	$\mathcal{K}_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$\mathcal{I}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1}{}_{\beta\alpha} == \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$-\frac{1}{2} \mathcal{K}_{\alpha\beta\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\beta\alpha\chi} n^{\chi} - \frac{1}{3} \eta_{\alpha\beta} \mathcal{K}_{\chi\delta}^{\delta} n^{\chi} +$ $\frac{1}{3} \mathcal{K}_{\chi\delta}^{\delta} n_{\alpha} n_{\beta} n^{\chi} + \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}_{2^+}^{\#1}{}_{\alpha\beta}$
$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$	$\mathcal{K}_{2^+}^{\#1}{}_{\beta\alpha\chi} == -\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$	$-\frac{1}{4} \mathcal{K}_{\alpha\beta\chi} + \frac{1}{4} \mathcal{K}_{\beta\alpha\chi} + \frac{1}{2} \mathcal{K}_{\chi\alpha\beta} - \frac{3}{8} \eta_{\beta\chi} \mathcal{K}_{\alpha\delta}^{\delta} + \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}_{\beta\delta}^{\delta} -$ $\frac{3}{8} \mathcal{K}_{\beta\delta}^{\delta} n_{\alpha} n_{\chi} + \frac{3}{8} \mathcal{K}_{\alpha\delta}^{\delta} n_{\beta} n_{\chi} + \frac{1}{4} \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\delta} +$ $\frac{1}{4} \mathcal{K}_{\delta\beta\chi} n_{\alpha} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{K}_{\delta\epsilon}^{\epsilon} n_{\alpha} n^{\delta} - \frac{1}{4} \mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\delta} -$ $\frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\delta} - \frac{1}{4} \mathcal{K}_{\delta\alpha\chi} n_{\beta} n^{\delta} - \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}_{\delta\epsilon}^{\epsilon} n_{\beta} n^{\delta} +$ $\frac{1}{4} \mathcal{K}_{\alpha\beta\delta} n_{\chi} n^{\delta} - \frac{1}{4} \mathcal{K}_{\beta\alpha\delta} n_{\chi} n^{\delta} - \frac{1}{2} \mathcal{K}_{\delta\alpha\beta} n_{\chi} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{K}_{\delta\alpha\epsilon} n^{\delta} n^{\epsilon} -$ $\frac{3}{8} \eta_{\alpha\chi} \mathcal{K}_{\delta\beta\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \mathcal{K}_{\delta\beta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \frac{3}{8} \mathcal{K}_{\delta\alpha\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon}$	$\mathcal{I}_{2^+}^{\#1}{}_{\alpha\beta\chi}$